

genaut — frontier report

source CSV : genaut/logs/1state2aplace/default.csv
corpus rows: 25

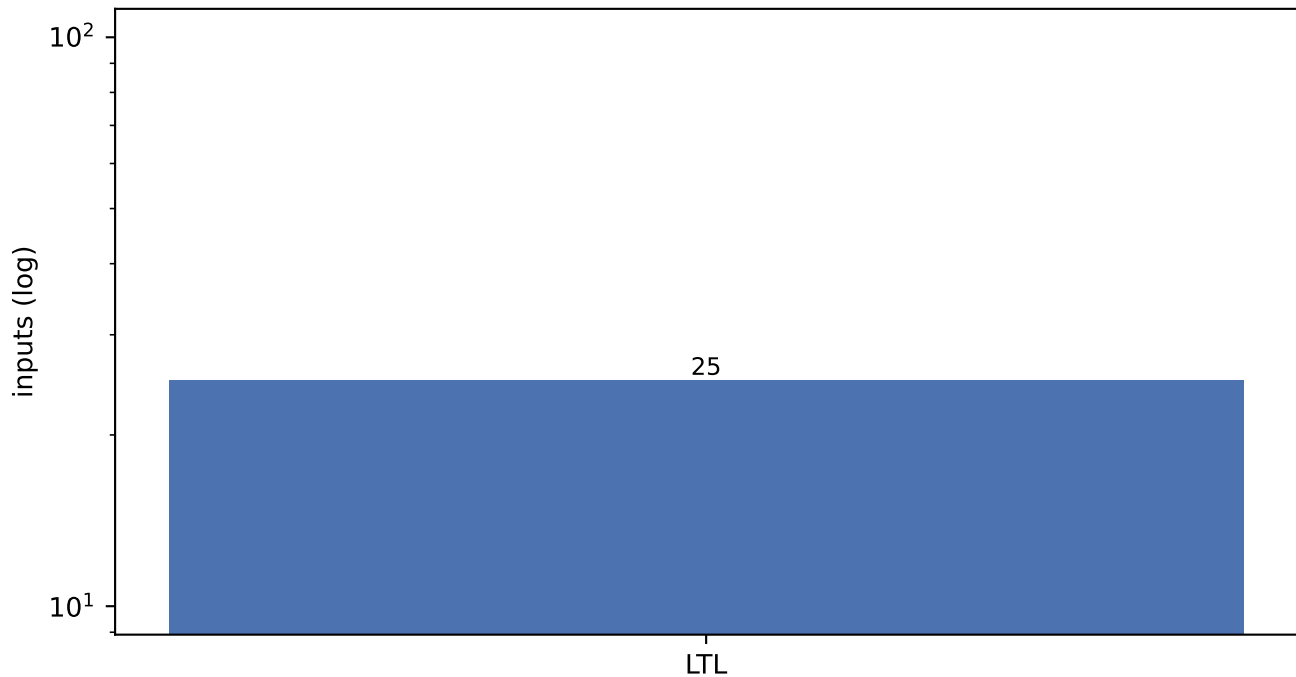
sections:

- Ventilation – LTL / not-LTL / ambiguous
- Formula collapse – structural (md5) concentration
- Size frontier – flattened formula size
- Routes to `true` – which technique built the universal 1
- Build time & timeouts
- Verification – spot-oracle validation

Ventilation — LTL / not-LTL / ambiguous

```
corpus inputs: 25
  LTL-definable (formula built) :    25  (100.0%)
    not-LTL (decided)           :      0  (  0.0%)
    ambiguous (timeout, unknown) :      0  (  0.0%)
decided (LTL or not-LTL): 25  (100.0%)
```

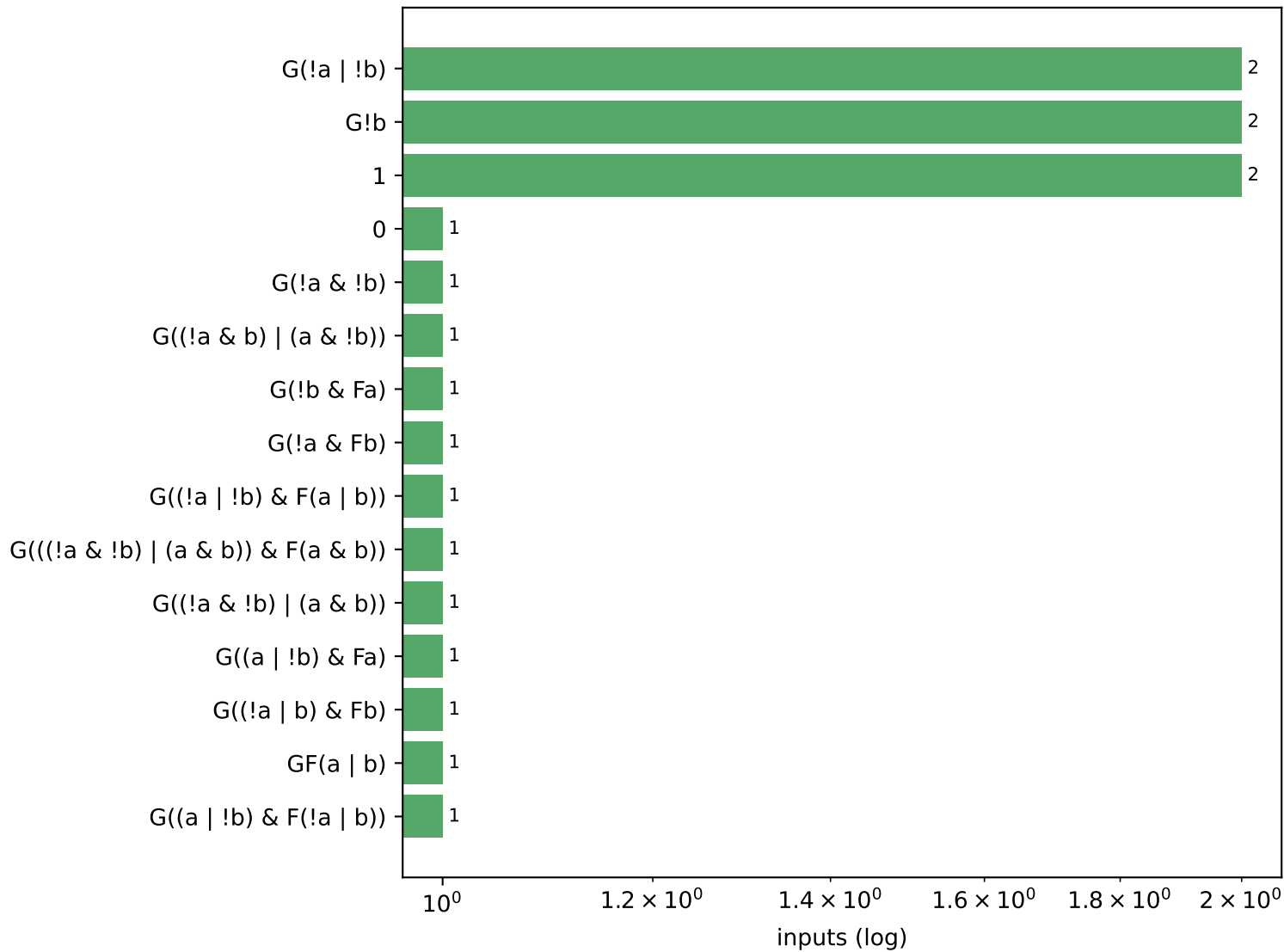
Answer frontier (by result)



Formula collapse — structural (md5) concentration

```
LTL answers: 25 (0 with a gated/unprintable string)
distinct structures (md5): 22
  80% of answers covered by the top 17 structure(s)
  90% of answers covered by the top 20 structure(s)
top 15 structures:
  2 ( 8.0%) G(!a | !b)
  2 ( 8.0%) G!b
  2 ( 8.0%) 1
  1 ( 4.0%) 0
  1 ( 4.0%) G(!a & !b)
  1 ( 4.0%) G(!a & b | (a & !b))
  1 ( 4.0%) G(!b & Fa)
  1 ( 4.0%) G(!a & Fb)
  1 ( 4.0%) G(!a | !b) & F(a | b))
  1 ( 4.0%) G(!a & !b | (a & b)) & F(a & b))
  1 ( 4.0%) G(!a & !b | (a & b))
  1 ( 4.0%) G(a | !b) & Fa)
  1 ( 4.0%) G(!a | b) & Fb)
  1 ( 4.0%) GF(a | b)
  1 ( 4.0%) G(a | !b) & F(!a | b))
```

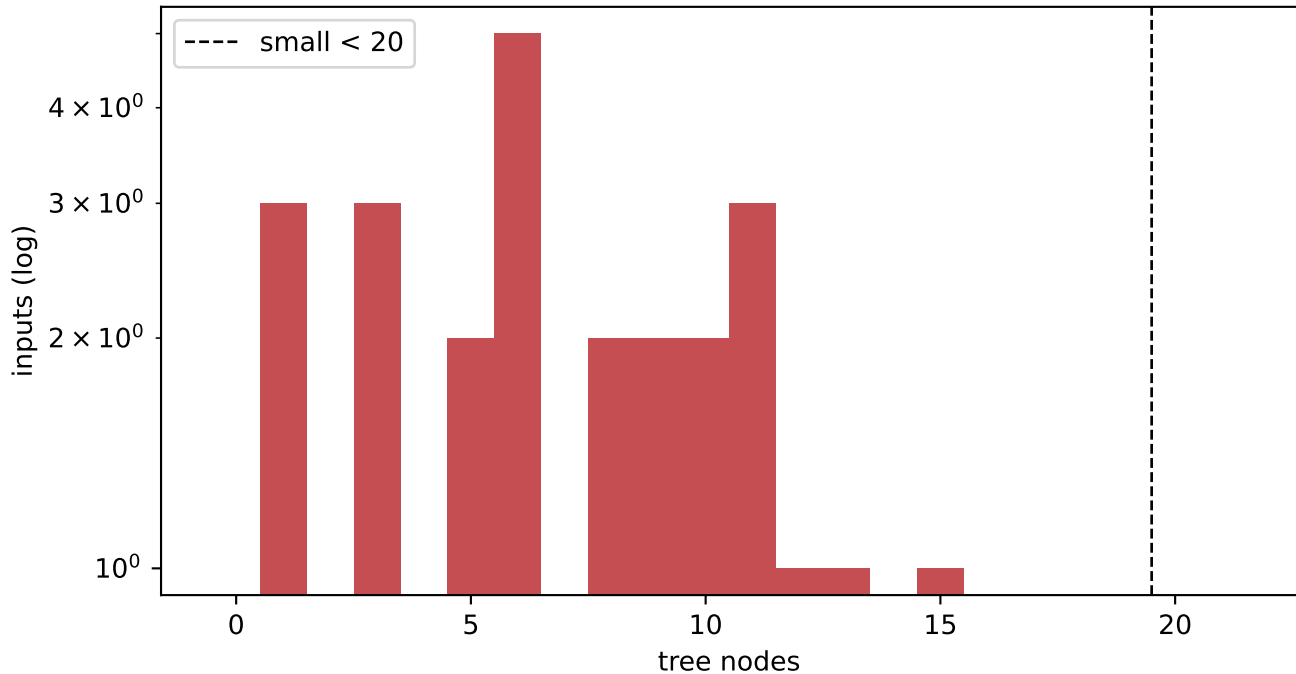
Top 15 reconstructed structures



Size frontier — flattened formula size

```
LTL answers with a size: 25
trivial (tree<=1, i.e. true/false): 3 ( 12.0%)
small   (tree<20):                  25 (100.0%)
larger  (tree>=20):                  0 (  0.0%)
tree nodes: median 6  mean 7.2  max 15
```

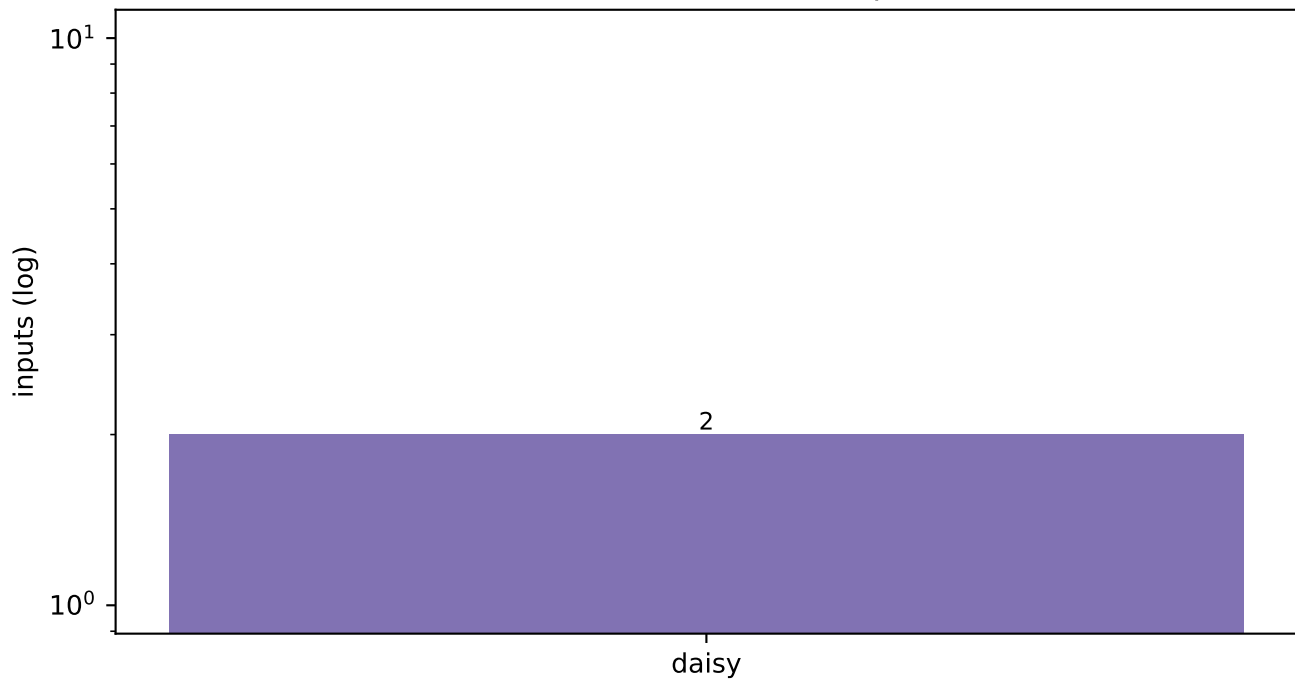
Flattened size (tree nodes); clipped at 21 (0 larger pooled at right)



Routes to `true` — which technique built the universal 1

```
universal answers (formula == '1'): 2  
  2 (100.0%) daisy  
build_s over true rows: total 0.1s  mean 0.073s  max 0.075s
```

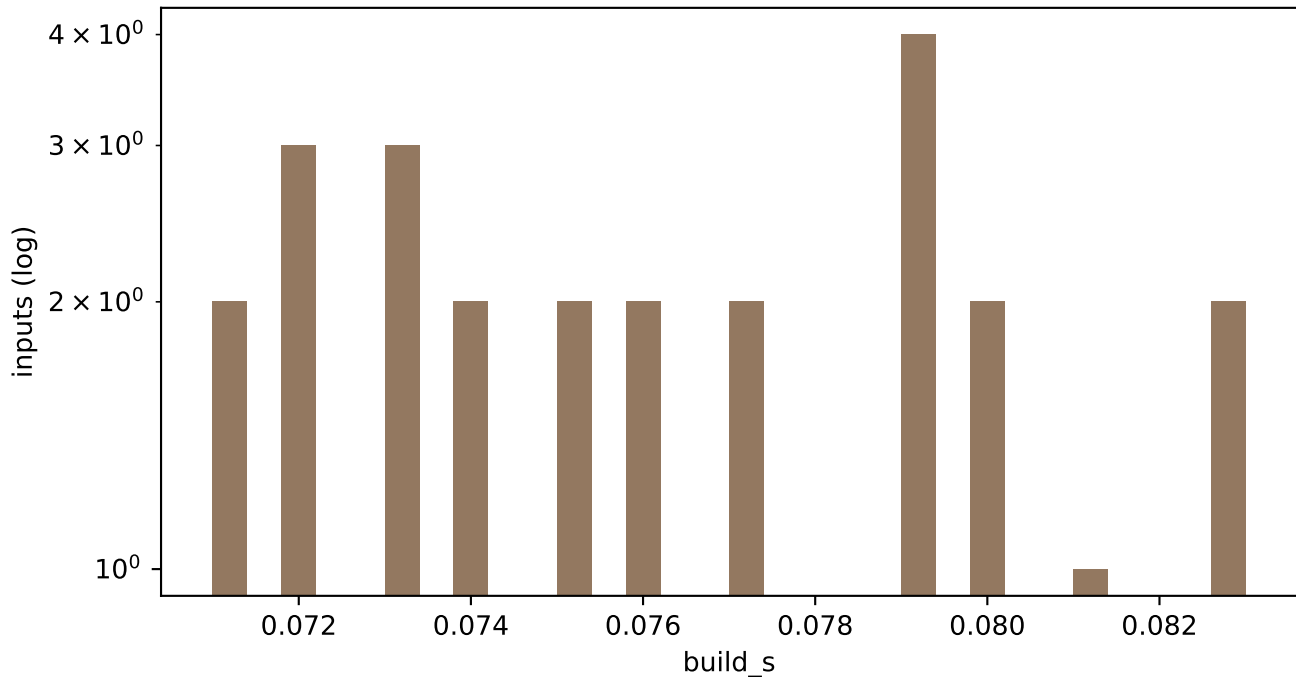

Routes to `true` (technique)



Build time & timeouts

```
total build: 1.9s    mean 0.076s    max 0.083s
  LTL          n= 25   mean 0.076s    max 0.083s
timeouts: 0
```

Build-time distribution



Verification — spot-oracle validation

validation over 25 LTL answers:
True 25 (100.0%)
NOT-equivalent (FALSE): 0 CLEAN